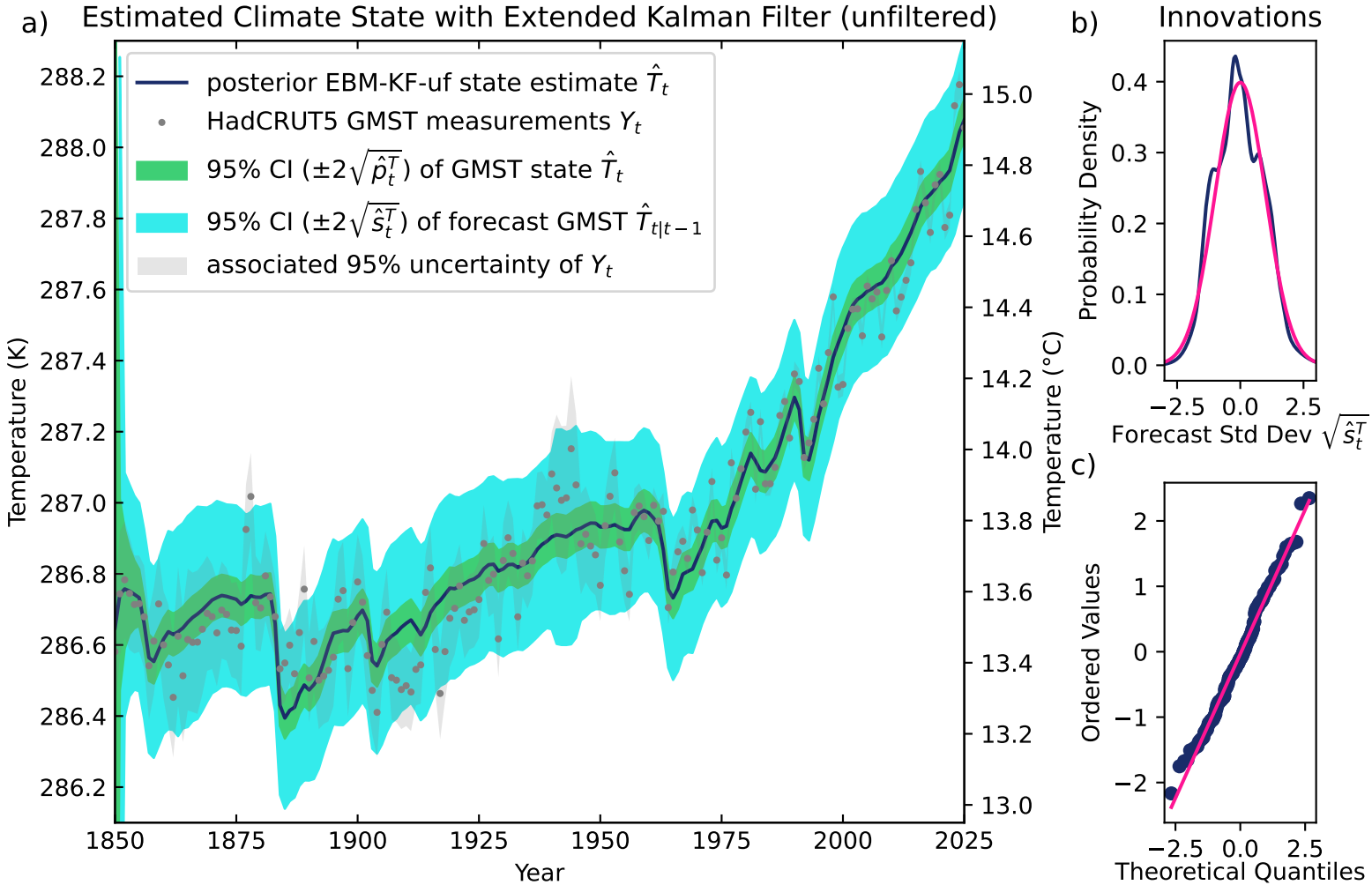
a) Estimated Climate State with Extended Kalman Filter (unfiltered)
posterior EBM-KF-uf state estimate $\hat{T}_t$
HadCRUT5 GMST measurements Y_t
95% CI ($\pm 2\sqrt{\hat{p}_t^T}$) of GMST state $\hat{T}_t$
95% CI ($\pm 2\sqrt{\hat{s}_t^T}$) of forecast GMST $\hat{T}_{t|t-1}$
associated 95% uncertainty of Y_t
Temperature (K)
Temperature (°C)
Year
288.2
288.0
287.8
287.6
287.4
287.2
287.0
286.8
286.6
286.4
286.2
15.0
14.8
14.6
14.4
14.2
14.0
13.8
13.6
13.4
13.2
13.0
1850
1875
1900
1925
1950
1975
2000
2025
b) Innovations
Probability Density
0.4
0.3
0.2
0.1
0.0
−2.5
0.0
2.5
Forecast Std Dev $\sqrt{\hat{s}_t^T}$
c)
Ordered Values
2
1
0
−1
−2
−2.5
0.0
2.5
Theoretical Quantiles